FACULTY OF COMPUTING
SEMESTER 2
SESSION 2023/2024

SECJ1023 SECTION 02

PROGRAMMING TECHNIQUE II

PROJECT PROPOSAL

LECTURER: DR. JOHANNA BINTI AHMAD

GROUP MEMBERS	MATRIC NO.
NAJMA SHAKIRAH BINTI SHAHRULZAMAN	A23CS0140
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
NABIL AFLAH BOO BINTI MOHD YOSUF BOO YONG CHONG	A23CS0252
AINA ZAFIRAH BINTI AHMAD MUZAMIR	A23CS8014

TABLE OF CONTENT

1 INTRODUCTION

The aim of this proposal is to suggest the development of an Object-Oriented Programming (OOP) based bus booking system; EasyBus: Simplifying Ticket Booking For You that uses C++ as the programming language. This system will make use of important OOP concepts like association, polymorphism, inheritance, encapsulation, and abstraction to manage the process of booking bus tickets and managing related operations. The bus booking system is designed to be a comprehensive system that tackles various challenges and inefficiencies that administrators and passengers have when it comes to managing and making bookings for bus tickets.

The proposed system will offer different functionalities for both passengers and administrators. Through a user-friendly interface, passengers will be able to browse available buses, choose preferred bus and seats, cancel booking and so on, while administrators will have the capability to manage bus routes, monitor bookings, and generate reports. Passengers will have access to a user-friendly interface where they can search for available buses, choose their desired routes, select seats, and complete the booking process efficiently. Meanwhile, administrators will have access to a dashboard where they can manage bus routes, update schedules, monitor bookings, and generate reports for analysis and decision-making purposes.

The goal in using C++ to develop the bus booking system is to take advantage of the efficiency, stability, and adaptability of the language in order to guarantee the system's smooth operation. The application of OOP concepts will improve the system's scalability and sustainability by facilitating flexible layout, code reuse, and easier maintenance.

Hence, the proposed bus booking system will serve as an essential tool in revolutionizing the bus ticket reservation process, offering both passengers and administrators a streamlined and efficient platform for managing bus bookings. It is hoped to provide a solid and dependable solution that prioritizes user experience and operational efficiency while addressing the changing needs of the transportation sector through the merging of OOP concepts and C++ programming.

2 SOLUTION

The proposed system would be developed using C++ programming language and will be using several functions that will deliver bus booking details for customers based on the data input. And it would also provide functions for the admin in handling the system.

The system will include classes such as Customer, Bus, and Add Ons. These classes will work together to provide a comprehensive and user-friendly system in booking the bus that will bring them to their desired destination.

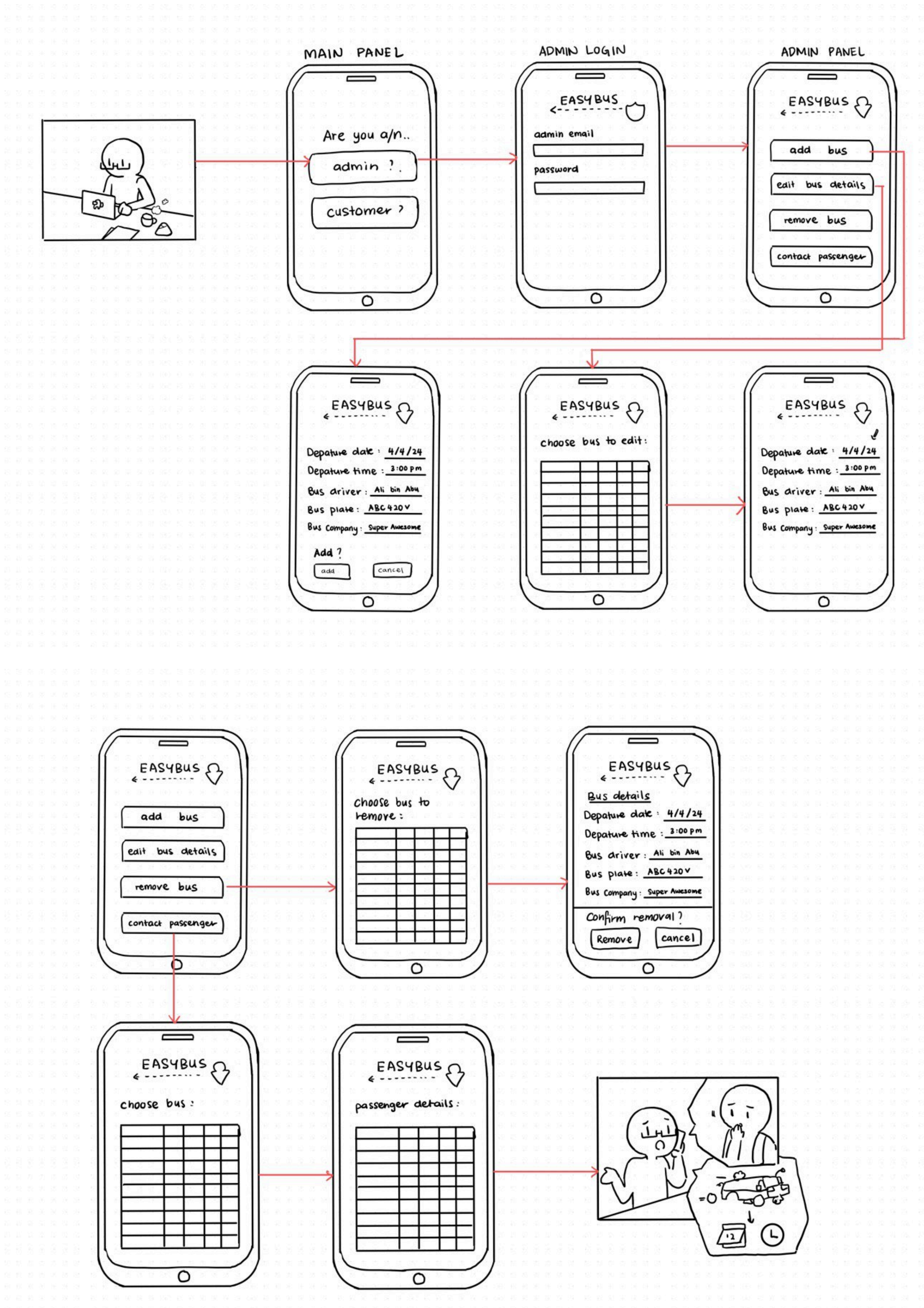
In the proposed system, the customer would be able to key in their personal details, such as name, age, gender, phone number and email. Next, customers would be brought to the next phase where they would enter their departure and return date, time they want to depart and return, which station they want to be picked up and where their final destination. After completing this stage, they would be asked whether to make some Add Ons or not such as snacks or drinks that they could eat while traveling. Lastly, when all the information has been filled in, customers will be brought to the payment phase, and after they are done with it, the system will email the receipt where it includes the payment and bus details as reference to the customer.

Furthermore, the classes for Bus and Customer would be working together for the admin system in managing the bus. In the admin's system, admin would be able to add buses in the system but the admin has to key in the bus details such as date, time, driver, bus plate number and from which company first. Next in the admin system, the admin could edit the bus details, remove buses that are not functioning or contributing, and get contact with the customers if there are some changes in the schedule or if any emergency happens.

3 CONCLUSION

In conclusion, our group proposed to implement an OOP-based system into our project, Bus Reservation System. We will apply OOP principles in this project to organize and construct our system in a way that makes it simple for us as programmers to understand and manage. For instance, we'll encapsulate our system within private and public accessible areas to prevent unauthorized parties from having direct access and make it difficult for them to hack into it. We will also apply additional OOP concepts, such as inheritance, polymorphism, and abstraction, to this system. As this system develops, bus driver management will become more efficient since it allows users to rate their experience and offers real-time visibility. Apart from that, C++ is the main programming language that we apply for this project, and to develop the systems, we will apply almost every aspect of our programming language knowledge. Taking advantage of our programming language experience of C++, we believe that the project will proceed smoothly and could be beneficial in the future for bus reservations.

4 STORYBOARD

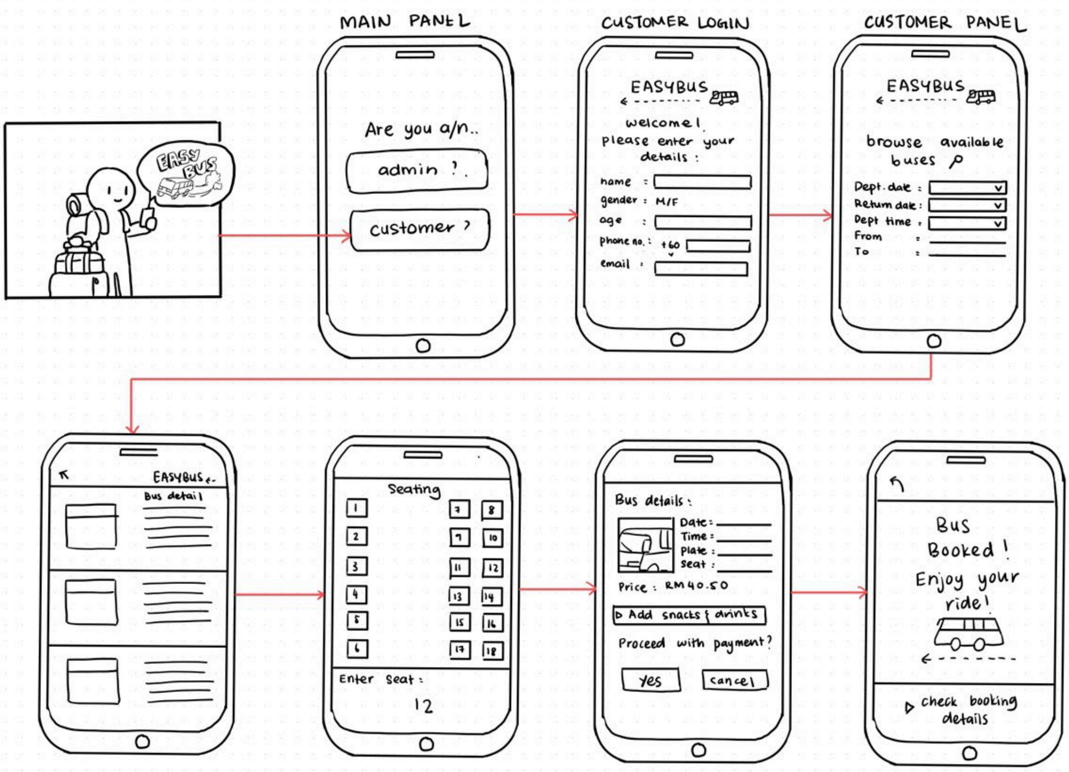


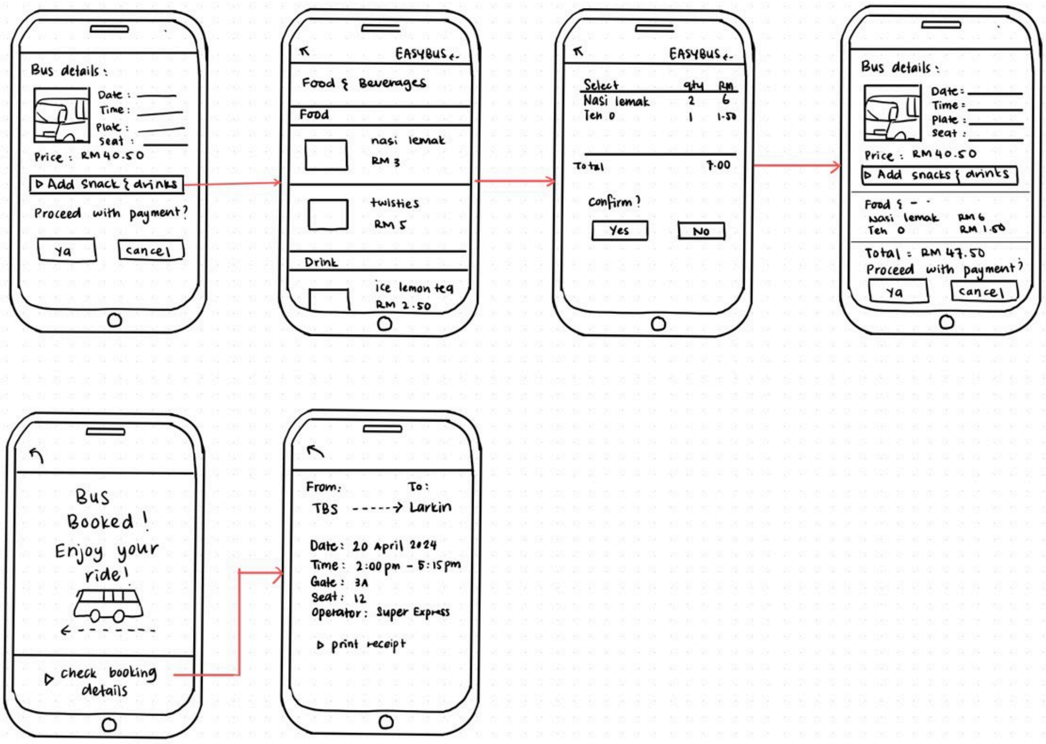
Bus details (for edit & remove):

Bus operators	Departure	Duration	Arrival	Plate	Driver	Fare
Super Awesome	11:00	3hr 15min	14:15	ABC1234	Ali	RM40.50
Blue Star	14:30	6hr	20:30	VDJ 3333	Abu	RM60.00
Bulan Restu	17:15	1hr 30min	18:45	CDA5413	Abi	RM35.50

Passenger details (for contact):

Name	Gender	Age	Phone no.	email	Seat	Date booked
Sara	F	25	012-3456789	sara@gm...	9	4/4/24
Joe	M	36	011-2233445	joe15@hotm...	15	29/3/24
Ahmad	M	21	019-8765432	ahmad@grad..	12	1/4/24





RECEIPT

EasyBus Booking Confirmation

Passenger info :
Name : Amirah binti Abdullah
ID no. : 030303 - 14 - 8888
Phone no. : 012 - 3456 7890
Payment status : Paid

Bus details :
From : Terminal Bersepadu Selatan
To : Larkin Terminal
Date & time : 20 April 2024 12.00 - 14.15
Bus no. : WPV 1132
Driver : Ali bin Abu
Seat no : 12
Terminal no. : 5
Bus operator : Super Express

Add-ons
Seat selection : 1
Meal : 1
Group discount : 15%

Payment Information :
Payment method : online banking
Total : RM 35.15